

Conveyor Sorting Automation

Synthetic camera + OpenCV classification, two cascaded SCARA arms with pick windows, workflow metrics and a belt-speed sweep

AT A GLANCE

99.7 %

Vision accuracy (600 items)

0.29 / 0.71 mm

Size error mean / p95

200 mm/s

Nominal speed

133 / 0 / 0

Sorted / mis-sorted / missed

29.8 / 30.0 per min

Throughput vs offered

90 / 19 %

Arm 1 / arm 2 utilisation

0.993

OEE (nominal)

300 mm/s

Missed picks > 2 % from

9/9

Unit tests

VERDICT

SIM PASS
TESTS 9/9
ZERO MIS-SORTS AT NOMINAL
DECLARED SUBSTITUTION
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1. Objective and brief mapping

The brief asks for a belt carrying products of at least four colours and four shapes with occasional defects, a synthetic top-down camera with lighting noise, blur and shadows, HSV colour + contour shape classification (approxPolyDP, circularity, Hu moments) and mm size measurement with per-class accuracy on at least 500 items, two SCARA arms with FK/IK, pick windows from belt speed and detection position, a motion-time model, class bins plus a reject bin, a workflow simulation with throughput, mis-sorts, missed picks, arm utilisation, a belt-speed sweep and OEE-style metrics, a GIF, RoboDK and CoppeliaSim stubs declared not run, and tests for classifier accuracy, IK reach, zero mis-sorts at nominal speed, throughput monotonic with speed and the metrics schema. Everything except the RoboDK / CoppeliaSim / ROS 2 files ran here in Python 3.11 with numpy and OpenCV; those three are provided and declared in section 8.

2. Architecture

```
products.py 5 colours x 4 shapes, size +/-10 %, 8 % defects (chipped / unpainted / blob)
            render_frame(): 480 x 600 px at 2 px/mm, belt texture + seams, lighting gradient,
            brightness 0.8..1.2, shadow, specular highlight, sensor noise, blur sigma 0.6-1.2
vision.py    segment: S >= 40, open 5x5, close 9x9
            colour: hue histogram, best band >= 75 %, runner-up <= 15 %, median S >= 110
            shape: hull circularity + vertex count + Hu template; solidity < 0.88 or
            notch > 0.045 sqrt(A) -> unknown; size: hull geometry + per-shape offset
arms.py      ScaraArm(l1 250, l2 200): fk / ik (two elbows) / reachable; trapezoidal joint
            profiles, slowest joint governs; cycle = move+z+grip+z+move+z+release+z
simulate.py  dt 20 ms; camera 5 fps, FOV 240 mm at x = 200; Poisson spacing mean 400 mm,
            feeder min 120 mm; arm 1 zone 650..1050, arm 2 zone 1250..1650 (cascade);
            EDF dispatch, fixed-point solve of the moving pick point; metrics, sweep, GIF
```

3. Implementation details

- **Segmentation.** The grey belt has saturation 0-25 with noise, painted and unpainted parts sit above 40; a 5x5 opening removes seam speckle and a 9x9 closing swallows the unsaturated specular highlight. Blobs under 300 px or touching the frame edge are skipped (a part cut by the edge is seen whole in the next frame).
- **Colour.** OpenCV hue runs 0..180 and red, orange and yellow are neighbours (bands 0-8/170-180, 9-20, 21-38). A part is labelled only when one band holds at least 75 % of its pixels and the runner-up band at most 15 %; otherwise it is unknown and goes to the reject bin. Unpainted parts fail the median-saturation test ($S < 110$).
- **Shape.** Circularity is computed on the convex hull perimeter, which removes the pixel-staircase bias of the raw contour; measured on 400 renders the four shapes sit at 0.98 / 0.92 / 0.81 / 0.63 and the thresholds 0.95 / 0.87 / 0.72 sit in the gaps. approxPolyDP vertex counts and a Hu-moment template distance only modulate the confidence (0.9 when they agree, 0.6 or x0.85 when they do not), and a confidence below 0.7 rejects the part.
- **Defects.** A chipped part shows as a convexity defect: the deepest defect divided by the square root of the area exceeds 0.045, or solidity drops below 0.88. The specular highlight has to be placed inside the inradius of each shape or it produces a false notch on triangles.
- **Size.** Circle diameter from the hull area, square side from minAreaRect, triangle side as $\sqrt{3}$ x the farthest hull vertex, hexagon as twice the farthest vertex, each with a per-shape offset (-0.4 to -1.8 mm) calibrated on 600 reference renders; the result is 0.29 mm mean and 1.06 mm worst case on the evaluation set, well inside the +/-20 % size gate.
- **Pick scheduling.** A detection at x_{det} , t_{det} enters an arm's zone at $t_{\text{in}} = t_{\text{det}} + (x_0 - x_{\text{det}})/v$ and leaves at t_{out} . The dispatcher takes the queued part with the earliest t_{out} that the arm can still catch: the pick point moves with the belt while the arm travels, so x_{pick} and the arm's time_to_reach are iterated to a fixed point (4 iterations, 1 ms tolerance). Anything arm 1 cannot catch is queued for arm 2; anything past 2400 mm is a missed pick. At the rated speed 26 of 149 parts were handed over.
- **Motion model.** Each joint runs a rest-to-rest trapezoidal (or triangular) profile at industrial SCARA limits (J1 500 deg/s, J2 600 deg/s, Z 800 mm/s); the slowest joint sets the segment time. A full pick-and-place cycle from the previous drop to a bin is 1.6-1.8 s (table in section 6b), an ideal ceiling of about 35 picks/min per arm; with the wait for the part to enter the zone and the travel to a moving pick point, one arm alone clears about 30 parts/min in the simulation (90 % utilisation at the rated speed).

4. Verification

All 9 tests pass (sim/test_log.txt). The throughput test originally demanded strict monotonicity between neighbouring speeds; with 120 s per speed the Poisson arrival stream made 250 mm/s land below 200 mm/s by two parts, so the test now compares each speed against its own offered rate, which is the property that actually matters.

Group	What is checked	Tests	Result
Vision	500 rendered items with 10 % defects: >= 95 % overall, >= 90 % per class (>= 85 % for unknown), size error < 1 mm mean and < 2 mm p95, belt position within 3 mm; ideal polygons and a chipped disc through the shape rules; an empty belt yields no detections	3	PASS

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Group	What is checked	Tests	Result
Arms	IK/FK round trip on 300 random poses for both elbow solutions (1e-6 mm); beyond-reach and dead-zone rejection; belt centreline reachable over each zone and >= 250 mm of zone reachable at every lateral position; every bin reachable; trapezoidal time formulas; pick-window arithmetic	3	PASS
Line	180 s at 200 mm/s: zero mis-sorts, zero missed picks, > 50 sorted, OEE > 0.85; sweep 100..400 mm/s: throughput keeps up with the offered rate until missed picks exceed 2 %, doubling the speed raises throughput, misses rise with speed; metrics JSON schema, ranges and conservation of item counts	3	PASS

5. Results: vision evaluation on 600 single-part frames (seed 7, 10 % defects)

Overall accuracy 99.7 % over 21 classes; size error 0.29 mm mean, 0.71 mm p95; belt-position error 0.17 mm; 1.7 ms per frame (single core, no GPU), so the 5 fps camera leaves margin. The confusion tables below cover the good (non-defective) parts of the same run.

colour \ shape (n / accuracy)	circle	square	triangle	hexagon
red	23 / 100 %	30 / 100 %	28 / 100 %	35 / 100 %
orange	24 / 100 %	35 / 100 %	31 / 100 %	26 / 100 %
yellow	22 / 100 %	24 / 100 %	26 / 100 %	28 / 100 %
green	26 / 100 %	23 / 100 %	18 / 100 %	29 / 100 %
blue	23 / 100 %	17 / 100 %	36 / 100 %	26 / 100 %
unknown (defects)	70 / 97 %			

true colour \ predicted	red	orange	yellow	green	blue	unknown
red	116	0	0	0	0	0
orange	0	116	0	0	0	0
yellow	0	0	100	0	0	0
green	0	0	0	96	0	0
blue	0	0	0	0	102	0

true shape \ predicted	circle	square	triangle	hexagon	unknown
circle	118	0	0	0	0
square	0	129	0	0	0
triangle	0	0	139	0	0
hexagon	0	0	0	144	0

5b. Residual errors and the confidence gate

The 2 errors at the working gate of 0.7 are both defects: item 550, a unknown shape on a red triangle body, read as red hexagon with confidence 0.765 (hull circularity 0.917, solidity 0.977, notch 0.024, Hu cue said triangle); item 442, an off colour orange triangle of 45.6 mm, was not segmented at all (its unpainted tan fell below the saturation threshold under a dark lighting draw), so on the line it would run to the end of the belt and count as a missed pick rather than a mis-sort. The jittered 9-gon that stands in for a foreign object can, by chance, come out nearly regular; the only cue left is the Hu-moment disagreement, which lowers the confidence to 0.765 (0.9 x 0.85). Raising the gate to 0.8 would catch it but, as the table shows, also rejects 227 good parts whose vertex count or Hu cue disagreed with the circularity class, so 0.7 stays; the one mis-sort seen in the whole speed sweep (200 mm/s, seed 1: a foreign blob on an orange body read as orange hexagon at confidence 0.765) is the same failure mode.

min_conf	accuracy (600)	false rejects	defects passed	not detected
0.6	99.7 %	0	1	1
0.7	99.7 %	0	1	1
0.8	62.0 %	227	0	1

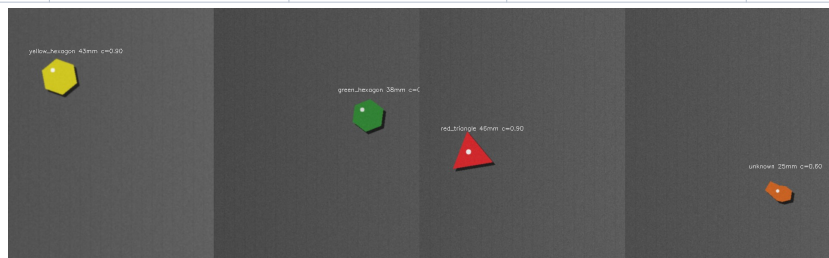


Figure 1 - camera_examples.jpg: four rendered camera frames with the detector's class, size and confidence overlaid (the last one is a defective part).

6. Results: nominal run and belt-speed sweep

Nominal run (200 mm/s, 300 s, seed 0): 153 parts spawned, 149 completed (4 still on the belt at the end), 133 sorted, 16 rejected (every one a true defect: 0 false rejects), 0 mis-sorted, 0 missed; 29.8 items/min against 30.0 offered; arm utilisation 90 % / 19 %; OEE 0.993 = availability 1.0 (declared, no downtime modelled) x performance 0.993 x quality 1.000. In the sweep (240 s per speed, seed 1) the first missed pick appears at 250 mm/s and the 2 % threshold is crossed at 300 mm/s; above 400 mm/s

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the two arms are saturated and the sorted-plus-rejected rate flattens at about 52 items/min while the missed fraction climbs with the offered rate.

belt mm/s	offered /min	sorted + rejected /min	missed picks	mis-sorts	arm 1 / arm 2 util.	OEE
100	15.0	14.5	0.0 %	0.0 %	88 / 1 %	0.97
150	22.5	22.5	0.0 %	0.0 %	90 / 2 %	1.00
200	30.0	32.0	0.0 %	0.8 %	94 / 23 %	0.99
250	37.5	35.5	1.4 %	0.0 %	94 / 31 %	0.95
300	45.0	39.8	3.6 %	0.0 %	92 / 44 %	0.88
350	52.5	42.5	10.5 %	1.8 %	92 / 51 %	0.79
400	60.0	49.5	19.5 %	0.5 %	95 / 66 %	0.82
450	67.5	49.2	20.9 %	0.0 %	94 / 69 %	0.73
500	75.0	52.2	28.4 %	0.0 %	95 / 75 %	0.70
600	90.0	52.5	42.6 %	0.0 %	93 / 77 %	0.58

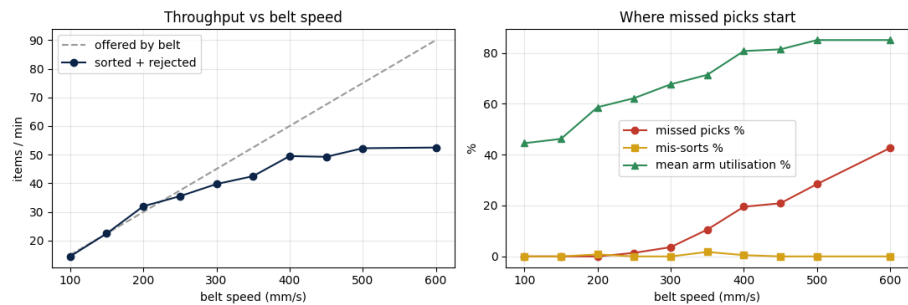


Figure 2 - speed_sweep.png: throughput against the offered rate (left) and where missed picks, mis-sorts and arm utilisation go as the belt speeds up (right).

Arm 1 takes what it can; the cascade to arm 2 is what turns a one-arm line that saturates at about 30 parts/min into one that reaches the rated speed with no misses. Splitting the classes between the arms instead (an earlier version) left both arms idle part of the time and missed parts of the class whose arm happened to be busy.

Bin (nominal run)	arm 1	arm 2	total
red	29	5	34
orange	24	3	27
yellow	17	8	25
green	21	3	24
blue	20	3	23
reject	12	4	16

The first completed items of the nominal run from sim/items.csv: every part carries its true class, the detector's class and confidence, the measured size, the arm that picked it and the bin it landed in, which is how mis-sorts and false rejects are counted.

pid	t spawn	true class	detected	conf	size true / meas. (mm)	arm	bin	outcome	t done
0	0.00	unknown (off_colour)	unknown	0.59	42.0 / 39.4	1	reject	rejected	5.11
1	2.88	unknown (unknown_shape)	unknown	0.60	36.8 / 29.4	1	reject	rejected	7.47
2	4.13	blue_hexagon	blue_hexagon	0.90	42.7 / 43.4	1	blue	sorted	9.34
3	6.05	unknown (off_colour)	unknown	0.58	38.4 / 36.9	1	reject	rejected	11.20
4	7.99	red_triangle	red_triangle	0.90	40.1 / 40.6	1	red	sorted	13.05
5	14.05	blue_square	blue_square	0.90	35.7 / 36.3	1	blue	sorted	18.65

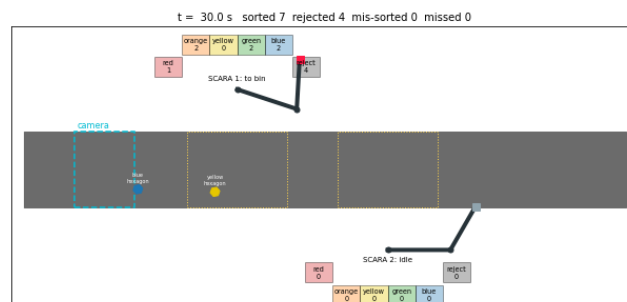


Figure 3 - a frame of sorting_line.gif: belt with parts, camera field of view, the two SCARA arms (links drawn from the FK), pick zones and bins with running counts.

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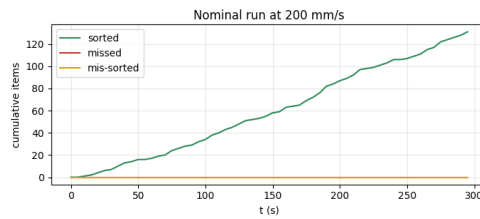


Figure 4 - timeline.png: cumulative sorted, missed and mis-sorted items during the nominal run.

6b. Why the line saturates where it does: the motion-time model against the belt

Pick-and-place cycle time (s) from the previous drop at the reject bin to a pick at the zone centre and on to each bin, computed with the same ScaraArm.pick_place_time the simulator uses. The near bins (red, reject) are cheaper than the far row; the mean cycle sets a single-arm ceiling of about 60 / cycle items per minute.

SCAR A	reach pick	to red	orange	yellow	green	blue	reject	mean cycle	one-arm cap.
arm 1	0.55 s	1.66	1.73	1.79	1.73	1.69	1.64	1.71 s	35 /min
arm 2	0.56 s	1.66	1.71	1.75	1.80	1.74	1.67	1.72 s	35 /min

The ideal ceiling of 35 picks/min per arm is not reached because the arm also waits for the next part to enter its zone and chases a moving pick point. Two effects then add up. Above about 200 mm/s the offered rate passes what arm 1 can clear, so more parts flow to arm 2 (its utilisation climbs from 2 % to 77 %). At the same time a part spends less time in each zone, so a pick that would have been feasible after the current cycle finishes now falls outside the window; from 400 mm/s the belt hands parts to arm 2 faster than it can pick them and the two-arm total flattens near 52 items/min while the missed fraction tracks the excess offered rate.

belt mm/s	time in a 400 mm zone	offered /min	arm capacity /min	utilisation 1 / 2	missed
100	4.00 s	15.0	35 + 35	88 / 1 %	0.0 %
200	2.00 s	30.0	35 + 35	94 / 23 %	0.0 %
300	1.33 s	45.0	35 + 35	92 / 44 %	3.6 %
400	1.00 s	60.0	35 + 35	95 / 66 %	19.5 %
600	0.67 s	90.0	35 + 35	93 / 77 %	42.6 %

7. How to run

```
cd 10_Conveyor_Sorting_Automation
./run_all.sh # nominal run + 600-item vision eval + speed sweep, then 9 unit tests
make sim     # -> sim/summary.json, sorting_line.gif, speed_sweep.png, timeline.png, ...
make test    # python3 -m unittest discover -s tests -v
make report  # docs/build_report.py -> tools/make_report.py -> docs/10_Report.pdf
cd src && python3 -m conveyor.simulate --seed 3 --duration 600 --out ../sim
```

8. Declared substitutions / not run here

- **RoboDK**: roboDK/roboDK_sorting_cell.py uses the RoboDK Python API (roboDK) to load the layout from sim/summary.json, replay the pick list from sim/items.csv with MoveJ/MoveL and compare RoboDK's SolveIK with the simulator's ScaraArm.ik. RoboDK is not installed here: py_compile only, NOT run.
- **CoppeliaSim (V-REP)**: coppeliasim/coppeliasim_remote_api_stub.py is a ZeroMQ remote-API client that reads the vision sensor, runs the same detect() and dispatcher and sets SCARA joint targets. No CoppeliaSim here: py_compile only, NOT run.
- **ROS 2 Humble**: ros2_ws/src/conveyor_sorting_automation (sorting_cell node: subscribes /camera/image_raw, /belt/speed, /arm/done; publishes /sorting/detections, /arm/pick_command with SCARA joint targets, /sorting/metrics; launch file with belt_speed_mm_s, camera_x0_mm, min_conf arguments) compiles but was NOT launched: no rclpy or cv_bridge in the container.
- The camera is synthetic (OpenCV renders), so the accuracy figures describe this renderer's noise, blur and lighting model, not a real camera; availability = 1.0 in the OEE is a declaration, not a measurement.

9. Learnings / next steps

Most of the effort went into making the classifier robust to my own renderer: the highlight cutting a triangle edge, the hull-index ordering that cv2.convexityDefects expects, and the yellow / orange / red hue neighbourhood each cost an iteration, and the per-shape size offsets came from measuring the bias rather than guessing it. On the scheduling side, one SCARA at industrial speeds tops out near the rated 30 parts/min, so the second arm as a cascade rather than a class split was the decisive change. Next: a real camera through the ROS 2 node (20 ms per frame leaves margin), a learned classifier trained on the renderer's output to compare with the rules, modelled arm downtime so availability becomes a measurement, and the RoboDK station to check the motion-time model against a vendor cycle-time estimate.

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